



Realising the potential impact of artificial intelligence for rare diseases – A framework

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ARTICLE INFO

Keywords:

Rare diseases
Patient journey
Generative artificial intelligence
Diagnosis
Care coordination

ABSTRACT

Rare diseases (RD) are conditions affecting fewer than 1 in 2000 persons, with over 7000 largely genetic RDs affecting 3.5 %–5.9 % of the global population, or approximately 262.9–446.2 million people. The substantial healthcare burden and costs, such as the \$1 trillion annual expense in the USA, highlight the urgent need for improved RD management. The International Rare Diseases Research Consortium (IRDiRC) addresses this need through global collaboration, aiming for timely and accurate diagnosis, development of 1000 new therapies, and methodologies to measure impact by 2027.

IRDiRC's initiatives include biannual meetings and workshops, like the AI-focused workshop in October 2023. This identified AI as crucial for advancing RD research and proposed a Framework for AI to enhance the RD patient journey by addressing efficiency and quality of life through modular solutions mapped to critical stages. The Framework integrates diverse data sources to improve diagnosis, treatment, and impact assessment, reflecting a holistic, cross-sector approach. By guiding multi-stakeholder efforts, the Framework aims to harness AI's potential to significantly improve rare disease care.

Introduction

Rare diseases (RD) are conditions with a specific pattern of clinical signs, symptoms, and findings that affect fewer than 1 in 2000 persons living in any World Health Organization-defined region of the world [1]. There are more than 7000 rare diseases, largely genetic (>80 %) in origin, with more than half affecting children [2,3]. Whilst rare diseases individually represent a small population burden, collectively they represent a much larger healthcare burden with between 3.5 % and 5.9 % of the world's population living with a rare disease, equating to an estimated 262.9–446.2 million people globally [3,4]. The direct and indirect costs of only a subset of rare diseases in the USA alone is approximately \$1 trillion p.a. and two thirds of national pediatric inpatient expenditure (105 billion p.a. RD versus 70 billion p.a. for

common diseases) [5–7]. Making rare diseases the biggest health cost. Recognizing the global unmet need of this grand health, and broader social, challenge has prompted the United Nations Convention on Addressing the Challenge of Persons Living with a Rare Disease (PLWRD) and Their Families, and inclusion in the Sustainable Development Goals. The unmet health needs of people living with RDs and their families include but are not limited to access to screening, diagnosis, therapy, physical and mental healthcare and support, and cross-sector care coordination and integration [4,8]. These require awareness, targeted interventions and pathways, including comprehensive education and training across the stakeholder spectrum.

Rare diseases are an exemplar domain for digital health [9,10], with multiple extant, and multifarious potential, AI applications [11–13]. The AI landscape has rapidly changed over the last few years, and

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<https://doi.org/10.1016/j.rare.2024.100057>

Received 2 July 2024; Received in revised form 25 November 2024; Accepted 19 December 2024

Available online 19 December 2024

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especially recently with the explosion of the accessibility and use of generative AI, with evolving breadth and depth use cases for rare diseases [14]. Given the rapid landscape evolution, there is a need for a rare diseases AI framework (Framework) to help harmonize and guide efforts. In line with the UN Resolution on Addressing the needs of the persons living with rare diseases and their families [15], such a Framework should traverse the breadth of the lived experience across all sectors of life. Herein we present the results of a workshop focused on progressing the topic under the auspices of International Rare Diseases Research Consortium (IRDIRC, <https://irdirc.org>).

IRDIRC unites national and international governmental and non-profit funding bodies, pharmaceutical and biopharmaceutical enterprises, medical device companies, umbrella patient advocacy organizations, and scientific researchers to foster global collaboration and harmonize the advance of rare diseases research. With global coverage spanning Africa, Asia, Australia, North and Latin America, and Europe, IRDiRC envisions enabling all individuals with rare diseases to receive an accurate diagnosis, care, and available therapy within one year of seeking medical attention. The vision is accompanied by three goals for 2027: (1) to improve and expedite rare disease diagnosis, (2) to contribute to the development of 1000 new therapies, focusing on rare diseases lacking approved treatments, and (3) to develop methodologies for measuring the impact of diagnoses and therapies on rare diseases patients. Task Forces, Working Groups and Workshops are instruments adopted by IRDiRC to address actionable topics identified by the Consortium in achieving its vision and goals.

Methods

In addition to frequent virtual meetings, IRDiRC has biannual face-to-face meetings of its constituents lead representatives (Consortium Assembly), Members of its Scientific and Constituent Committees and operational Executive (Operating Committee). These serve to help guide and prioritize IRDiRC focusses. At the March 2023 meeting in Berlin, artificial intelligence was raised up simultaneously by multiple IRDiRC committees as a critical area of increasing opportunity to advance rare diseases research against the IRDiRC goals. Over the coming months, committee members worked together to deepen and refine a cross-committee proposal, to progress towards a Framework for AI for rare diseases.

A workshop focused on the topic was held at a hybrid meeting (face-to-face and virtual) held in Montreal Canada in October 2023. The general aim was to document the areas where the latest advances in AI can contribute to the RD patient journey and propose a modular framework that maps critical questions to various stages of the journey. This would then enable solution-builders to provide AI-driven answers

to specific questions while intrinsically contributing to the improving the patient's experience in the overall journey.

Following a presentation highlighting key definitions and selected examples of current AI approaches being deployed against the 3 IRDiRC goals, the workshop discussions advanced through various stages, mapped to a natural progression from the RD patient journey to a unified framework for AI in RD:

The starting point was represented by the RD patient journey, as per the schema used by the Data, Digital and Technology Taskforce of the World Health Organization (WHO) Global Network for Rare Diseases to scope solutions and identify technology needs for the Network – Fig. 1. The participants have immediately recognized that in virtually all cases (diagnosed or not) care is the most important factor with relevance to the quality of life of RD patients and families. A holistic and cross-sector care coordination model – with individual sectors including, for example, Physical and Mental Health, Education, Disability and Social Services – was identified, as delivered by the Rare Care Centre in Western Australia [16] – Fig. 2.

1. Critical questions and areas of development were then identified with a focus predominantly on equitable access to care and improving the quality of life of RD patients and families.
2. Finally, a unified framework was proposed – with care as a central piece – by mapping the questions identified above to AI components emerged from the most recent wave of large language models (LLMs) [17–19].

Results

The schema depicted in Fig. 1 presents the RD patient journey as a joint partnership with clinical professionals. From a clinical perspective, this journey can be conceptually divided into pre-diagnosis and post-diagnosis phases (i.e., Complex clinical presentation and Diagnosis versus the subsequent steps), leading to efficiency-driven questions (see Q1 and Q2 below). However, from a patient's perspective, it highlights the need for a cross-cutting focus on equity and improving quality of life (see Q3 below). The model of care depicted in Fig. 2, and implemented increasingly around the globe, provides the answer to this latter transversal by implementing a cross-sector care coordination approach – hence recognizing health (as a dimension of life) as an entry point rather the means to an end. The questions then formulated by the workshop participants naturally captured the role AI will play in facilitating advances in the various areas of the RD patient journey when focusing on the overall quality of life, and not just the health dimension.

Below we list the three groups of questions that emerged from the discussions.

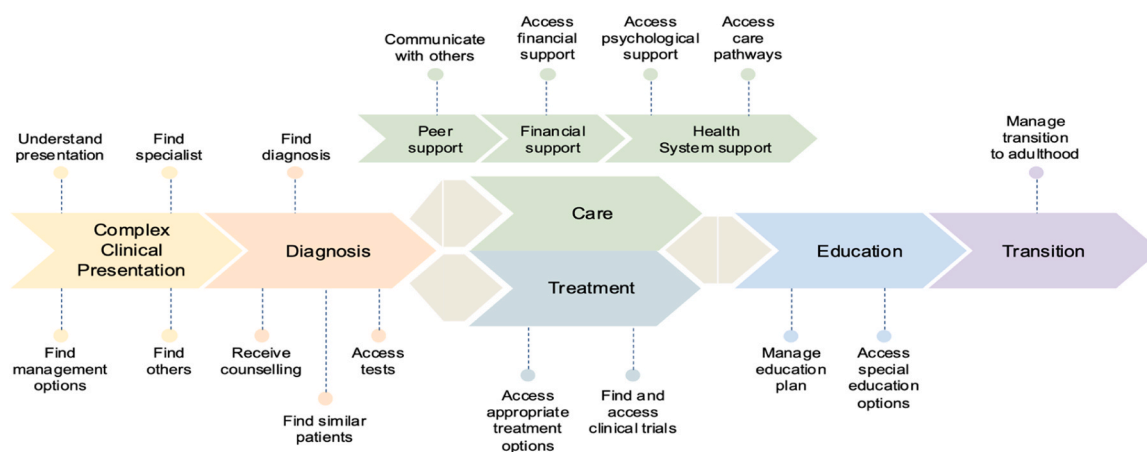


Fig. 1. The visual schema of the data, digital and technology taskforce of the world health organization global network for rare disease. The schema is based on patient and family journey from seeking a diagnosis to care and its coordination.

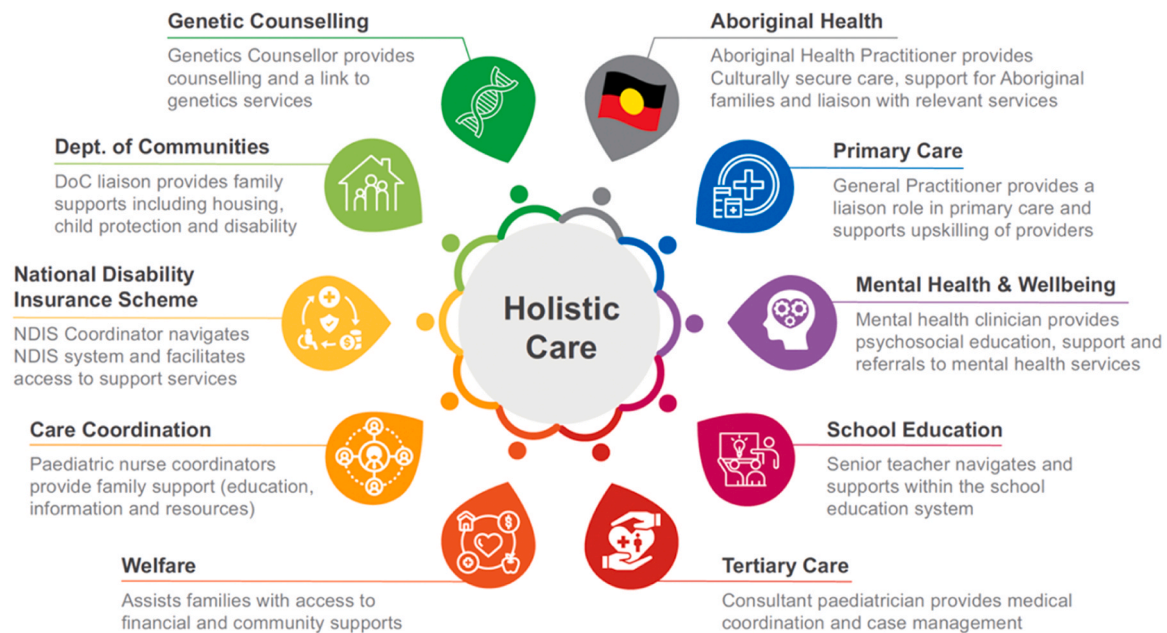


Fig. 2. Rare care centre model of care - a world first transformative, co-designed cross-sector Model of Care, to provide holistic care and support in an integrated and responsive service, and hence put choice and control back into the hands of families.

Q: How can AI reduce the diagnostic odyssey?

A: Using generative models with appropriate contextual data points from clinical presentation through diagnosis and care/treatment (either via fine-tuning or via retrieval augmented generation) could lead to greater efficiencies and understanding of disease, hence reducing the diagnostic odyssey. More concretely, it can contribute to:

- **Improving diagnostic methods** – generative models can identify subtle patterns and anomalies in medical data that traditional methods might miss, leading to more accurate and early detection. LLMs can also generate synthetic data to augment limited patient data, enhancing the robustness of diagnostic tools.
- **Improving disease knowledge at molecular level** – generative AI can predict the structure and function of complex biomolecules, revealing novel insights into disease mechanisms. Additionally, these models can simulate genetic mutations and their impact on molecular interactions, aiding in the discovery of potential biomarkers and therapeutic targets.
- **Finding and interpreting functional studies** – LLMs can predict which experimental setups and conditions (including the type of model organism) are most likely to yield meaningful results, thus streamlining research efforts. Furthermore, generative models can analyze and synthesize data from various studies, providing a comprehensive interpretation that highlights critical diagnostic markers and biological pathways.

Q: How can AI improve care coordination? (with and without diagnosis)

A: We can use AI to improve rare diseases care coordination in several ways, such as:

- **Drug discovery, development and repurposing** – generative models can accelerate drug discovery and development by predicting potential drug candidates and their interactions with disease-specific molecular targets, thereby reducing the time and cost of experimental screening. Additionally, they can identify existing drugs that may be repurposed for rare diseases by simulating their effects on relevant

biological pathways and suggesting modifications to enhance efficacy.

- **Clinical trials access, matching and modeling** – LLMs can predict the most suitable trials based on individual genetic and clinical profiles, thus enhancing patient recruitment and participation. Moreover, these models can simulate various trial scenarios, optimizing study designs and predicting outcomes to ensure more efficient and effective trials.
- **Public health interventions** – generative models can be used to predict the spread and impact of diseases within populations, allowing for more targeted and efficient resource allocation. LLMs can also simulate the potential outcomes of various intervention strategies, aiding in the development of effective policies and programs tailored to the needs of rare disease communities.
- **Care coordination models** – AI can help evaluate the effectiveness of different models of care coordination for rare conditions based on the taxonomy of care coordination dimensions and by taking into account the phenotypic profiles of the patients instead of relying solely on an existing diagnosis.

Q: How can AI empower people and families living with rare diseases and improve the quality of life?

A: AI can be used to empower, educate and support patients and families living with rare diseases by providing them with personalized information, guidance, and resources in several ways, such as:

- **Disease education** – AI can help provide patients and families with customized information and resources about rare diseases, such as causes, diagnosis, treatment, research, support groups, etc. This can help them learn more about their condition, cope with their challenges, and advocate for their needs.
- **Symptom matching** – generative models can help match the self-reported symptoms of patients and families with those of rare diseases and provide them with relevant information and suggestions. This can help them understand their condition better, seek appropriate care, and connect with other patients and families.
- **Digital therapeutics** – AI can help deliver personalized interventions and treatments for rare diseases through digital platforms, such as

apps, games, or wearable devices. These can help patients and families manage their symptoms, improve their outcomes, and monitor their progress.

- **Support groups** – LLMs can analyze patient data and match individuals with groups that share similar symptoms or conditions, even in the absence of a formal diagnosis. These models can also identify emerging patterns in patient experiences, facilitating the creation of new support networks tailored to specific needs and challenges faced by rare disease patients.

The resulting framework is depicted in Fig. 3. Through the Framework, a holistic and cross-sector lens can be applied across the patient and family journey to scaffold and underpin the conceptualisation of AI approaches for rare diseases. Some examples, set against the three IRDiRC goals (diagnosis, treatment and impact assessment) are as follows:

- **Diagnosis.** The Framework can be applied to cross-sector approaches to addressing the diagnostic odyssey of rare diseases. Specifically, data from various sectors, not just health data, could be used to identify indicators (red-flags) for rare diseases. For instance, in the Education sector - excessive school absence, visits to the school nurse, frequent bullying [20,21], requirement for special education, a combination of these and other factors; alone or in combinations with access patterns of community, financial and social services might be identified as red flags. These red flags could be in turn presented in lay language, and/ or translated into multiple languages using AI and used to support enhanced referrals. Further, data could be combined compared and contrasted between administrative, clinician ascertained and patient reported data.
- **Treatment and Impact Assessment.** Combinations of data, models, ontologies and AI could be used to both generate novel approaches and assess the outcomes and impact of care coordination. For instance, data ascertained directly from patients and using lay language and clinical layers applications of the human phenotype ontology [22], could be combined to holistic (e.g. Indigenous [23, 24]) models of care, nursing care ontologies [25] and social determinants of health (e.g. through the Gravity Project [26]) to unlock novel insights into, generate new models for, and assess impact of care coordination.

Artificial intelligence is progressing at unprecedented speed and

scale, including for applications to rare disease. Given this, and the breadth and depth of unmet rare disease need, it is timely to deliver a framework to help identify gaps and opportunities, and guide and harmonize the considerable multi-stakeholder efforts needed, to realize the full potential of AI for improving the lives of people and families living with rare disease.

Discussion

The IRDiRC Consortium Assembly (Assembly) and scientific committees represents all members of IRDiRC and as such embodies a globally unique collective knowledge, thought leadership and insight base that benefits from a diversity of perspectives that traverses the multistakeholder spectrum of rare diseases research. The process described herein unlocked the power of the IRDiRC Consortium Assembly through a workshop to ideate on use cases of AI within and across sectors and through-out the patient journey, to develop a Framework for AI in rare diseases. The intent was not to provide a comprehensive assessment of the state-of-play, nor to detail in depth the individual challenges and solutions. But rather to iterate upon previous organizational discussions and thinking in the context of a rapidly moving field. Specifically, to provide a conceptual Framework to help guide the diverse range of further studies required, in a manner that mirrors the whole-of-life and cross-sector needs of PLWRD and their families.

A successful implementation of the overall framework, or part of thereof, will depend on a series of criteria, including:

- **Communication** – multiple channels of communication will have to be used to provide support for the patients, in addition to easy-to-understand information.
- **Personalization** – the degree to which the application of the framework will be tailored to the specific needs of the individual patients.
- **Optimization** – the use of data from multiple channels in an integrated fashion, and yet remaining compliant possibly across different geographies.
- **Compliance** – with the regulations, regulatory standards and legal frameworks in each geography of application.
- **Engagement** – patients and the clinical stakeholders require a shared set of values and desired outcomes, which can only be achieved via continuous communication and connection.

Furthermore, we see the need for the following necessary steps to

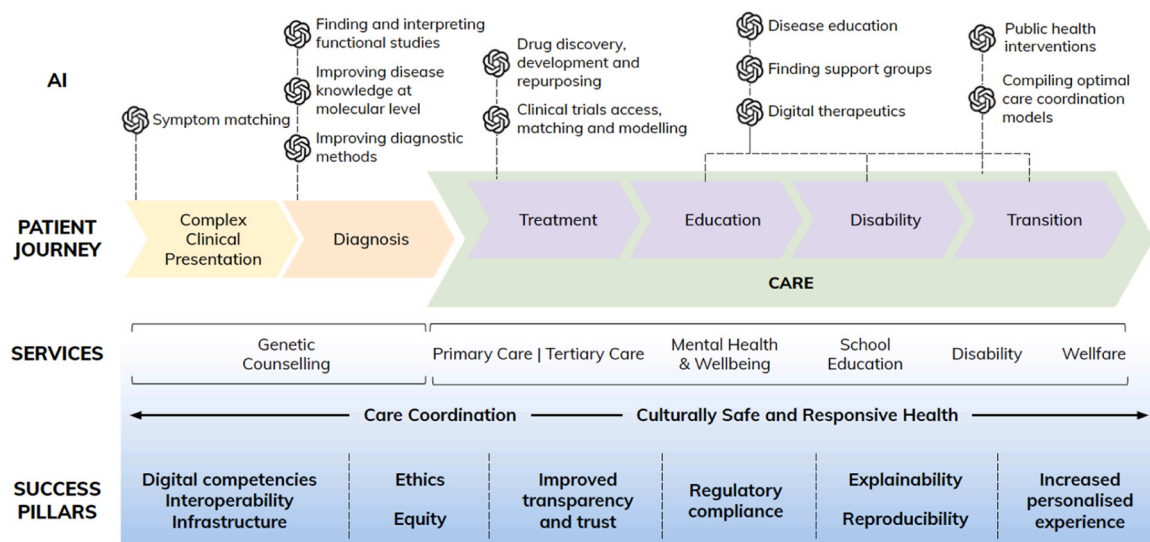


Fig. 3. Unified Framework for AI in Rare Diseases, emerged from the RD patient journey and focusing on care. Key aspects of the journey are mapped to AI components emerged from the most recent wave of Large Language Models.

be either performed, or at least appropriately planned, to achieve long term sustainability of any implementation:

- Investment in digital competencies, interoperability and infrastructure: to ensure that diverse healthcare systems and technologies can communicate effectively, share data seamlessly, and operate efficiently together. Robust digital skills and a well-connected infrastructure are crucial for fully harnessing AI's potential to enhance diagnostics, personalize treatment, and improve patient outcomes.
- Improved transparency and trust in technology and data: to gain the confidence of patients and healthcare providers, ensuring that AI-driven decisions are reliable, human-interpretable and ethical [27]. This fosters an environment where data can be securely shared and utilized, enhancing the accuracy and effectiveness of AI applications in healthcare. Ultimately, this trust is critical for the widespread adoption and success of AI technologies in medical settings.
- An increasingly personalized experience: to allow AI to tailor diagnostics and treatments to the unique genetic and clinical profiles of individual patients – e.g., DeepGestalt that identifies genetic syndromes from facial images [28]. This precision enhances the effectiveness of care and can lead to better outcomes in conditions that are often poorly understood and treated. Personalization also helps to identify subtle variations and patterns that may be missed with a one-size-fits-all approach, driving advancements in rare disease management.
- A collaborative relationships with regulators: to ensure that AI applications in the rare diseases field meet stringent safety and efficacy standards. This cooperation facilitates the approval process, ensures compliance with ethical guidelines, and builds trust among stakeholders. Additionally, working closely with regulators helps to navigate complex regulatory landscapes, enabling faster and more efficient deployment of innovative AI solutions. A notable initiative exemplifying this is the U.S. Food and Drug Administration's (FDA) precisionFDA platform (<https://precision.fda.gov/>). This cloud-based, open-source platform fosters collaboration among researchers, clinicians, and regulators to advance precision medicine. By providing a secure environment for sharing data, tools, and methodologies, precisionFDA facilitates the development and validation of AI applications, ensuring they meet stringent safety and efficacy standards. This cooperation streamlines the approval process, ensures compliance with ethical guidelines, and builds trust among stakeholders, thereby accelerating the deployment of innovative AI solutions in the rare diseases field.

There are several projects and initiatives that cover some of these steps, developed predominantly by the tech giants:

- IBM Watson for Oncology is an AI platform designed to assist oncologists in making personalized treatment decisions for cancer patients. It analyzes patient data against a vast database of medical literature, clinical guidelines, and treatment outcomes to provide evidence-based treatment recommendations [29].
- DeepMind, in partnership with Moorfields Eye Hospital in London, developed an AI system capable of diagnosing eye diseases from retinal scans. The AI uses deep learning to detect conditions like age-related macular degeneration and diabetic retinopathy [30].
- Project InnerEye is an AI initiative by Microsoft Research that focuses on developing tools for the automatic analysis of 3D medical images. It assists in radiotherapy planning by accurately segmenting tumors and healthy tissues in imaging data [31].

Limitations and cautions

Using the latest wave of AI developments in healthcare, in general, comes with a series of limitations and cautions that are critical to be considered and have been documented in the literature.

Firstly, a clear distinction should be made between utilizing generative AI models for hypothesis generation vs employing them for direct patient or clinician guidance. In hypothesis generation, the

primary obstacle is the scarcity of comprehensive datasets, especially concerning rare diseases, which limits the AI's ability to identify meaningful patterns and generate reliable hypotheses. Conversely, providing direct guidance involves ensuring the accuracy and reliability of AI-generated information, maintaining patient privacy, and seamlessly integrating AI tools into clinical workflows. Moreover, the interpretability of AI outputs is crucial; clinicians must understand and trust the AI's reasoning to make informed decisions. These challenges underscore the need for careful planning, execution, and management of expectations in adopting generative AI [32]. Additionally, the potential for AI to misinterpret rare disease presentations due to limited training data necessitates rigorous validation and oversight to prevent misdiagnoses and ensure patient safety [33].

Secondly, applications like ChatGPT or Gemini have shown significant potential in healthcare, but their tendency to produce hallucinations – fabricated or incorrect information – poses serious challenges. For example, an AI might misinterpret patient symptoms and suggest a non-existent medical condition, leading to misdiagnosis. A concrete instance is when AI-generated medical advice includes incorrect medication dosages, which could result in harmful patient outcomes. Such inaccuracies and hallucinations have been highlighted repeatedly in the literature – e.g., [34] – as well as the need to address them to enable a safe integration of the technologies in clinical practice [35].

Thirdly, maintaining a human in the loop when using generative AI in healthcare is essential due to the risk discussed above. Clinical oversight and/or human supervision was emphasized by multiple studies [36,37] as a direct result of the potential inaccurate medical advice and to prevent the dissemination of erroneous information.

Finally, and with an utmost importance for the rare diseases domain, achieving equitable AI faces significant challenges due to generative AI hallucinations that can introduce or amplify biases. For example, an AI diagnostic tool trained predominantly on images of lighter-skinned individuals may misclassify skin conditions in patients with darker skin tones, leading to misdiagnosis or delayed treatment. This exacerbates existing health disparities by providing less accurate care to underrepresented groups – e.g., the analysis performed in [38] on an algorithm that exhibited significant racial bias, affecting millions of patients by allocating fewer resources to Black patients than to White patients with the same level of need. Biases in AI models can lead to inequitable healthcare outcomes, which emphasizes the need for diverse datasets and fairness-aware algorithms to mitigate these issues [39].

Conclusion

The IRDiRC Consortium Assembly, with its diverse, multi-stakeholder insights, utilized a workshop to ideate on AI use cases across sectors and the patient journey, developing a conceptual Framework for AI in rare diseases. This Framework aims to guide future studies, reflecting the comprehensive needs of people living with rare diseases and their families. It provides a holistic, cross-sector approach to conceptualize AI applications for rare diseases, addressing diagnosis, treatment, and impact assessment. Moreover, it suggests using diverse data sources, such as education and social services, to identify diagnostic indicators and enhance referrals. Finally, it proposes integrating patient-reported data with clinical and administrative data to develop novel care models and assess their impact. This timely framework aims to guide multi-stakeholder efforts to harness AI's potential in improving rare disease care.

Ethics statement

N/A

CRediT authorship contribution statement

Gareth Baynam: Writing – review & editing, Writing – original

draft, Methodology, Conceptualization. **David A Pearce:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Chun-Hung Chan:** Writing – review & editing, Writing – original draft, Methodology. **Tudor Groza:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

GB acknowledges that he is funded through the Angela Wright Bennett Foundation, The McCusker Charitable Foundation via Chanel 7 Telethon Trust, Mineral Resources Limited, the Stan Perron Charitable Foundation and the Perth Children's Hospital Foundation.

Data availability

No data was used for the research described in the article.

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